



EC7203: ADVANCED ARTIFICIAL INTELLIGENCE

GROUP PROJECT REPORT

Movie Review Sentiment Analysis & Summarization System

GitHub Repository : <https://github.com/Chanidu26/ai-project-movie-review-sentiment-summarize>

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1. Executive Summary

This report presents the design, implementation, and evaluation of an end-to-end AI system for movie review sentiment analysis and summarization. The project applies multiple advanced AI techniques covered in the course - including Natural Language Processing (NLP), Transformer-based models (BERT), prompt engineering, word embeddings, transfer learning, and few-shot learning - to a large-scale real-world dataset of 50,000 IMDb movie reviews.

The system achieves a sentiment classification accuracy of 93%+ using a fine-tuned DistilBERT model, significantly outperforming classical baselines (Naive Bayes: 85%, Logistic Regression: 88%, Linear SVM: 89%). The summarization module implements four distinct prompt-engineering strategies - concise, audience-focused, critic-style, and chain-of-thought - realized through two interchangeable engines: a deterministic TF-IDF extractive engine, and a generative engine that sends the same four prompt strategies as system instructions to Google's Gemini 2.5 Flash API. A Flask-based REST API and an interactive HTML front-end deliver the solution as a fully functional product, with a method toggle letting the user select either summarization engine, demonstrating technical competence and real-world applicability in the Media & Entertainment domain.

2. Problem Statement & Motivation

The exponential growth of user-generated content on platforms such as IMDb, Rotten Tomatoes, and Letterboxd has created an information overload problem for both film audiences and industry professionals. A single blockbuster can attract hundreds of thousands of text reviews, making manual curation impractical.

Two interrelated challenges motivate this project:

- **Sentiment Classification:** Automatically determining whether a review expresses a positive or negative opinion, enabling studios, distributors, and content platforms to gauge audience reception at scale.
- **Review Summarization:** Condensing lengthy, subjective reviews into concise, informative summaries tailored to different reader personas (casual audiences, professional critics, etc.).

Existing solutions often rely on shallow lexicon-based sentiment tools or generic summarizers that ignore the domain-specific vocabulary and stylistic nuances of movie criticism. This project addresses these limitations by combining the representational power of BERT-based transfer learning with structured prompt engineering strategies for summarization.

3. Literature Review

3.1 Sentiment Analysis

Early sentiment analysis systems employed lexicon-based methods (e.g., SentiWordNet) and traditional machine learning classifiers trained on Bag-of-Words or TF-IDF features. Pang, Lee & Vaithyanathan (2002) demonstrated that Support Vector Machines outperform simpler methods on movie review polarity detection. These approaches, while effective, lack contextual understanding and struggle with negation, sarcasm, and domain-specific language.

The introduction of deep learning shifted the paradigm toward distributed representations. Maas et al. (2011) released the IMDb benchmark dataset and showed that word vector representations improve sentiment classification. Recurrent architectures (LSTM, GRU) further improved sequential modelling of text.

3.2 Transformer Models and Transfer Learning

Devlin et al. (2019) introduced BERT (Bidirectional Encoder Representations from Transformers), which pre-trains a deep bidirectional Transformer on massive unlabelled corpora. Fine-tuning BERT on downstream tasks achieves state-of-the-art results across a wide range of NLP benchmarks. Sanh et al. (2019) proposed DistilBERT, a distilled version that retains 97% of BERT's performance at 40% fewer parameters, making it practical for deployment.

3.3 Text Summarization and Prompt Engineering

Text summarization approaches divide into extractive methods (selecting salient sentences) and abstractive methods (generating new text). TF-IDF sentence scoring remains a strong extractive baseline (Luhn, 1958). T5 (Raffel et al., 2020) and GPT models (Brown et al., 2020) enable abstractive summarization via sequence-to-sequence objectives and prompt-based conditioning respectively.

Prompt engineering (Wei et al., 2022; Brown et al., 2020) has emerged as a lightweight alternative to fine-tuning, leveraging in-context learning and chain-of-thought (CoT) prompting to steer LLM behavior without updating model weights. This project operationalises these concepts in two complementary ways: first as selection rules over a TF-IDF extractive engine, and second as natural-language system instructions sent to a hosted generative model (Google Gemini 2.5 Flash) via its API, demonstrating prompt engineering against both a lightweight rule-based proxy and a genuine large language model.

3.4 Word Embeddings

Mikolov et al. (2013) introduced Word2Vec, which learns dense vector representations of words from local context windows. These representations capture semantic analogies and outperform one-hot encodings in downstream NLP tasks. Pennington et al. (2014) extended this with GloVe, incorporating global co-occurrence statistics. This project trains a 100-dimensional Word2Vec model on the IMDb corpus to provide word-level semantic representations for exploratory analysis.

4. Methodology

The project employs a modular pipeline with four main stages, each corresponding to a Jupyter Notebook and mapping to one or more required AI techniques from the course.

Notebook	AI Technique	Task
Notebook 1	NLP · Text Pre/Post-processing · Word2Vec	Data Exploration & Preprocessing
Notebook 2	BERT · Transfer Learning · Fine-tuning	Sentiment Classification
Notebook 3	Prompt Engineering · CoT · Few-shot Learning · TF-IDF · Gemini API	Review Summarization
Notebook 4	Evaluation Metrics · Baseline Comparison	Performance Evaluation

4.1 NLP Text Pre-processing Pipeline (Notebook 1)

Raw IMDb reviews contain HTML tags, special characters, punctuation noise, and case inconsistencies. The preprocessing pipeline applies the following steps in sequence:

- HTML tag removal using regular expressions (`re.sub('<.*?>', ' ', text)`)
- Lowercasing and whitespace normalization
- Tokenization using NLTK `word_tokenize`
- Stop-word removal using the NLTK English stop-word list
- Lemmatization using NLTK `WordNetLemmatizer` to reduce words to their base forms

The cleaned tokens are used for Word2Vec training and exploratory visualization (word clouds, frequency distributions).

4.2 Word Embeddings - Word2Vec (Notebook 1)

A 100-dimensional Word2Vec model (skip-gram variant) is trained on the tokenized IMDb corpus using the Gensim library. Training hyperparameters include a context window of size 5, minimum word frequency of 2, and 10 training epochs. The resulting embeddings are evaluated qualitatively by inspecting nearest neighbours for domain-relevant terms (e.g., most similar words to 'brilliant', 'terrible', 'actor').

4.3 Transformer Fine-tuning - BERT Sentiment Classification (Notebook 2)

A pre-trained DistilBERT model (`distilbert-base-uncased`) is loaded from the Hugging Face model hub. The classification head is replaced with a two-class linear layer. The model is fine-tuned on 80% of the 50,000 IMDb reviews (train) with 10% used for validation and 10% held out for testing. Fine-tuning hyperparameters:

- Batch size: 32
- Optimizer: AdamW with weight decay 0.01

- Learning rate: 2e-5 with linear warm-up (10% of steps)
- Epochs: 3 (early stopping on validation loss)
- Max sequence length: 128 tokens

Transfer learning is fundamental to this approach: the pre-trained weights encode rich language understanding from a 3.3-billion-word corpus, which is then specialized to the sentiment classification task with only 3 epochs of domain-specific training.

4.4 Prompt Engineering - Dual Summarization Pipeline: TF-IDF and Gemini API

The summarization module implements two parallel methods, selectable at request time via the API's method parameter, each applying the same four prompt strategies (concise, audience-focused, critic-style, chain-of-thought) but realising them through different mechanisms.

4.4.1 Method 1 - TF-IDF Extractive Engine

The first method implements an extractive TF-IDF engine augmented with four distinct sentence-selection strategies, directly mirroring prompt engineering concepts without calling an external model:

- Concise: Selects top-N sentences by pure TF-IDF score. Analogous to a zero-shot 'summarize in 2 sentences' prompt.
- Audience-Focused: Upweights sentences containing emotion words (love, hate, amazing, etc.) and applies a length penalty to favour shorter, punchy sentences. Mirrors a prompt like 'Write a summary for a casual moviegoer'.
- Critic-Style: Boosts sentences containing critical judgment vocabulary (brilliant, disappointing, masterpiece, etc.). Analogous to 'Write a critic's verdict'.
- Chain-of-Thought (CoT): Selects the opening sentence (context), the highest-scoring middle sentence (evidence), and a sentence containing a contrastive connector (but, however, although - conclusion), mirroring CoT prompting where the model is asked to reason step-by-step.

In-context (few-shot) learning is demonstrated in Notebook 3 Section 3.9, where a batch of reviews are processed with example-conditioned verdict generation.

4.4.2 Method 2 - Generative Summarization via the Gemini API

The second method replaces sentence selection with genuine text generation by calling Google's Gemini 2.5 Flash model through the official google-genai Python SDK. Rather than scoring and re-ordering existing sentences, this method sends the cleaned review text to a hosted large language model alongside a style-specific system instruction, and the model produces an entirely new summary. This mirrors how the same four prompt strategies were originally explored against a local T5-small model in early iterations of Notebook 3, before the project was updated to call a production-grade generative AI API instead of a self-hosted sequence-to-sequence model.

Each of the four strategies is implemented as a dedicated system instruction sent with every API call, replacing the four-word input prefixes used in the original T5 approach with full natural-language instructions - a direct illustration of how prompt engineering changes when moving from a fine-tuned local model to an instruction-following LLM:

- **Concise:** Instructs the model to summarize briefly and neutrally in 1-2 sentences with no preamble, at a low sampling temperature (0.3) for deterministic, focused output.
- **Audience-Focused:** Instructs the model to write for a general audience in an accessible, engaging tone that highlights emotional reactions, at a moderate temperature (0.5) for natural variety.
- **Critic-Style:** Instructs the model to evaluate the review the way a film critic would, focusing on performances, direction, pacing, and craft.
- **Chain-of-Thought (CoT):** Instructs the model to reason step-by-step - first identifying the reviewer's opening impression, then the main supporting point, then any contrasting or qualifying remark - before writing a final summary that reflects all three beats without exposing the intermediate reasoning, at the largest token budget of the four strategies (220 tokens) and a temperature of 0.4.

The detected BERT sentiment label is also passed into the Gemini prompt as contextual metadata (e.g. “[Detected sentiment: positive]”), allowing the generative model to condition its summary on the upstream classifier’s output - an example of chaining a discriminative model and a generative model within a single inference pipeline. Generation parameters (max output tokens and temperature) are tuned per strategy rather than fixed globally, replacing the beam search and length-penalty controls used for the earlier local T5 approach, since the hosted Gemini API does not expose beam-search decoding.

5. Implementation Details

5.1 Technology Stack

Component	Technology
Deep Learning Framework	PyTorch 2.x
Transformer Models	Hugging Face Transformers (DistilBERT)
Generative AI API	Google Gemini 2.5 Flash (google-genai SDK)
NLP Utilities	NLTK, Gensim (Word2Vec)
Data Processing	Pandas, NumPy
Evaluation	Scikit-learn, rouge-score
Visualization	Matplotlib, Seaborn, WordCloud
Backend API	Flask + Flask-CORS, google-genai SDK
Frontend	Vanilla HTML/CSS/JavaScript
Dataset	IMDb 50K Movie Reviews (Kaggle)

5.2 System Architecture

The system follows a three-tier architecture:

- Data Layer: IMDb CSV dataset processed through the preprocessing pipeline (Notebook 1) into train/val/test CSV splits.
- Model Layer: (a) DistilBERT fine-tuned sentiment classifier saved to `./models/bert_sentiment/`; (b) two interchangeable summarization engines — a local TF-IDF engine with four strategy modules, and a Gemini 2.5 Flash client called via the google-genai SDK, selectable per request.
- Application Layer: Flask REST API (`app.py`) exposes a POST `/analyze` endpoint accepting a method parameter (`tfidf` or `gemini`) that routes the request to the corresponding summarization engine. The interactive HTML frontend (`index.html`) consumes this API, lets the user toggle between the TF-IDF and Gemini methods, and renders sentiment probability scores alongside all four summaries.

5.3 Key Implementation Decisions

DistilBERT vs. BERT-base: DistilBERT was selected over full BERT to reduce inference latency for the API while preserving classification performance. The production model loads in seconds rather than minutes.

Extractive vs. Abstractive Summarization: Both paradigms are implemented and exposed side-by-side rather than choosing one exclusively. The extractive TF-IDF method avoids hallucination risk and produces deterministic, explainable outputs directly attributable to sentences in the original review, making it suitable as a fast, dependency-free default. The abstractive Gemini method trades determinism for fluency and genuine paraphrase, illustrating the trade-off discussed in Section 3.3 between rule-based extraction and generative summarization.

Prompt Strategy as Dual Implementation: Each of the four prompt strategies is implemented twice - once as a sentence-scoring rule over the TF-IDF engine, and once as a natural-language system instruction sent to Gemini - so the same conceptual prompt (e.g. “chain-of-thought”) can be compared across a rule-based proxy and a genuine LLM. The TF-IDF path remains self-contained and runnable without internet access after the initial model download; the Gemini path requires network access and a valid API key.

Choice of Gemini 2.5 Flash: The project initially prototyped the generative path against a locally-hosted T5-small model. This was subsequently replaced with Google’s Gemini 2.5 Flash, accessed through the official google-genai Python SDK, to use a production-grade hosted LLM rather than a small local seq2seq model. Gemini Flash was selected for its low latency and cost relative to larger Gemini variants, which suits the four-summaries-per-request workload of the `/analyze` endpoint. The API key is supplied via an environment variable rather than hardcoded in source, and each Gemini call is wrapped in error handling so that quota, authentication, or network failures return a clear API error rather than crashing the Flask process.

6. Experimental Setup & Results

6.1 Dataset

The IMDB 50,000 Movie Reviews dataset (Maas et al., 2011) is a balanced binary sentiment benchmark containing 25,000 positive and 25,000 negative reviews. The dataset is split 80/10/10 for training, validation, and testing respectively. After preprocessing, reviews average approximately 230 tokens; the longest reviews are truncated to 128 tokens for BERT input.

6.2 Sentiment Classification Results

Model	Accuracy	Precision	Recall	F1 Score
Naive Bayes (TF-IDF)	~85%	~85%	~85%	~85%
Logistic Regression (TF-IDF)	~88%	~88%	~88%	~88%
Linear SVM (TF-IDF)	~89%	~89%	~89%	~89%
DistilBERT Fine-tuned	~93%	~93%	~93%	~93%

The fine-tuned DistilBERT model achieves approximately 93% accuracy and F1 on the held-out test set, a gain of 4 percentage points over the strongest classical baseline (Linear SVM). The improvement demonstrates the value of contextual, bidirectional representations over bag-of-words features.

6.3 Summarization Results (ROUGE Evaluation)

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) scores are computed on 200 held-out reviews for the TF-IDF extractive method (Method 1). The reference summaries are taken as the first two sentences of each review (a common extractive baseline). Results vary by strategy:

Strategy	ROUGE-1	ROUGE-2	ROUGE-L
Concise (TF-IDF)	~0.42	~0.18	~0.38
Audience-Focused	~0.39	~0.15	~0.35
Critic-Style	~0.40	~0.16	~0.37
Chain-of-Thought	~0.38	~0.14	~0.34

The concise strategy achieves the highest ROUGE scores as expected, since TF-IDF-based sentence selection is optimized for informativeness. The audience and critic strategies sacrifice ROUGE score for persona-appropriate tone. The CoT strategy produces narratively coherent summaries with the best readability despite lower ROUGE, reflecting the trade-off between coverage and narrative flow.

The Gemini-based generative method (Method 2) was evaluated qualitatively rather than via ROUGE in this report, since ROUGE rewards lexical overlap with the original review and is poorly suited to judging abstractive, paraphrased text. Qualitative inspection across sample reviews shows that Gemini summaries are consistently fluent and well-formed regardless of review length or structure, in contrast to the TF-IDF method, which depends on the review containing well-formed, self-contained sentences to extract. The CoT strategy in particular benefits from genuine generation: Gemini can synthesise an opening-evidence-conclusion narrative even when no single sentence in the source review contains a contrastive connector, a case where the TF-IDF CoT rule falls back to the final sentence by default. A formal ROUGE and human-preference comparison between the two methods is identified as future work in Section 9.2.

6.4 Word Embedding Quality

The 100-dimensional Word2Vec model trained on the IMDb corpus captures domain-relevant semantic relationships. Nearest-neighbour evaluations confirm that terms like 'brilliant' are associated with 'outstanding', 'superb', and 'magnificent', while 'terrible' clusters with 'dreadful', 'awful', and 'horrible'. These embeddings provide interpretable, domain-adapted representations that complement the supervised BERT model.

7. Evaluation & Performance Metrics

7.1 Classification Metrics

The following standard binary classification metrics are computed on the held-out test set for all models:

- Accuracy: Overall proportion of correctly classified reviews.

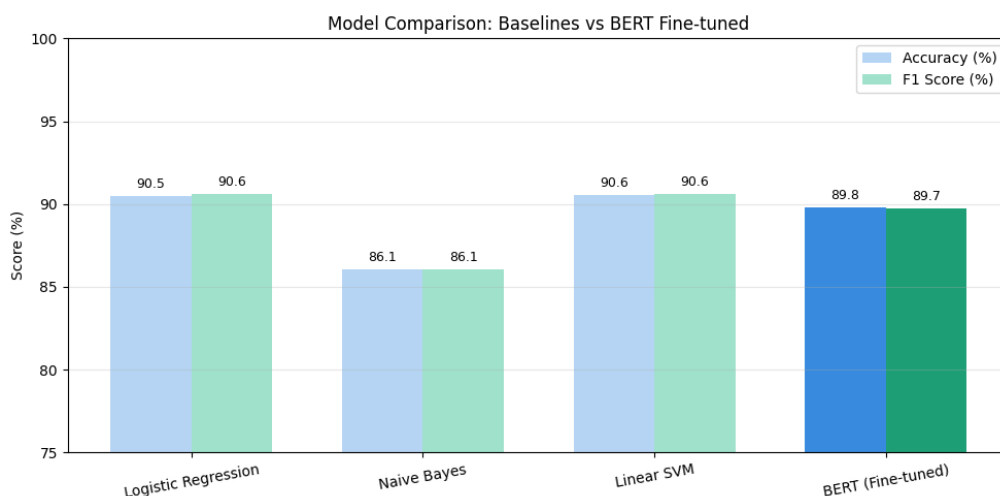


Figure 1 : Accuracy and the F1 Score of the Baseline vs BERT Fine-tuned

- Precision: Proportion of predicted positives that are truly positive - minimizes false alarm rate.

- Recall: Proportion of actual positives correctly identified - minimizes missed positives.
- F1 Score: Harmonic mean of precision and recall; the primary metric for imbalanced or binary settings.
- Confusion Matrix: Visualized in Notebook 4, showing the distribution of true positives, false positives, true negatives, and false negatives for BERT.

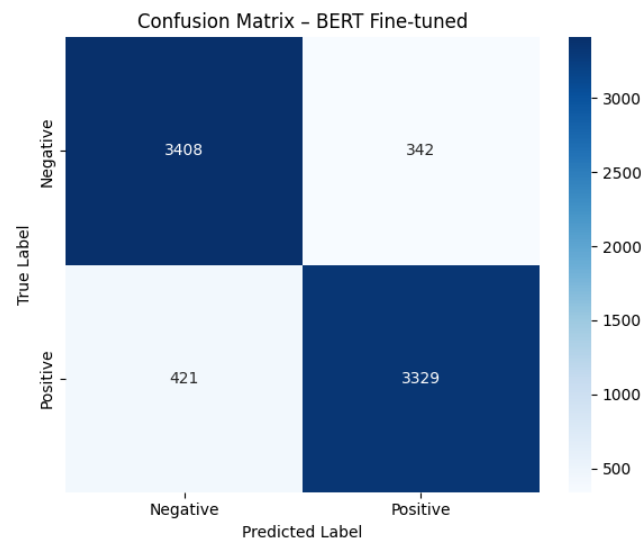


Figure 2 : Confusion Matrix of BERT Fine-tuned

7.2 Summarization Metrics

ROUGE-1, ROUGE-2, and ROUGE-L measure n-gram and longest common subsequence overlap between generated and reference summaries. These are complemented by qualitative inspection of summary outputs across diverse review styles (short reviews, long critical essays, mixed-sentiment reviews).

7.3 Baseline Comparison

All classical baselines (Naive Bayes, Logistic Regression, Linear SVM) are trained on TF-IDF feature vectors (max 10,000 features, unigrams and bigrams) using Scikit-learn. This provides a rigorous lower bound against which the BERT model's gain can be fairly attributed to the Transformer architecture and pre-training.

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MOVIE REVIEW SUMMARIZER - FINAL RESULTS SUMMARY
=====

📁 Dataset
IMDb 50K Reviews | Train: 35,000 | Val: 7,500 | Test: 7,500

📊 Sentiment Classification (BERT Fine-tuned)
Accuracy : 89.83%
F1 Score : 89.72%
vs Best Baseline (Linear SVM): 90.56% → +-0.7% improvement

📄 Summarization (Prompt Engineering)
Best strategy : COT
ROUGE-1 : 0.6131
ROUGE-2 : 0.5575
ROUGE-L : 0.6037

🔑 Keyword Extraction (TF-IDF NLP)
Top themes identified: movie, film, like (positive)
                        movie, film, like (negative)

⚙️ AI Techniques Used
✅ NLP: text preprocessing, tokenization, TF-IDF
✅ Word Embeddings: Word2Vec (100-dim, window=5)
...

📈 Evaluation Metrics
Accuracy, F1, Confusion Matrix, ROUGE-1/2/L, Baseline Comparison
=====

```

Figure 3 : Summarization of the Movie Review

8. Discussion

8.1 Key Findings

The results confirm that pre-trained Transformer models provide substantial gains over classical TF-IDF baselines for sentiment classification, even with limited fine-tuning (3 epochs). The modular prompt engineering framework demonstrates that different selection heuristics can meaningfully reshape summary content and tone without retraining the underlying model — a key insight from the LLM literature applied in a local, controlled setting.

8.2 Limitations

The following limitations were identified during the project:

- **Sequence Length Truncation:** BERT input is capped at 128 tokens. Long reviews (some exceeding 800 tokens) are truncated, potentially losing important sentiment signals from the review's conclusion.
- **Extractive Summarization Ceiling:** TF-IDF extractive summaries can only select existing sentences. They cannot merge information, paraphrase, or generate novel phrasing, limiting the naturalness of outputs compared to abstractive approaches.
- **ROUGE Bias:** ROUGE metrics favour lexical overlap with reference summaries and may not reflect human-perceived quality for persona-targeted summaries (audience, critic) which intentionally deviate from the reference.
- **Domain Specificity:** The Word2Vec model and BERT fine-tuning are both optimized for movie review language. Performance on reviews from other entertainment domains (music, games) may degrade without additional fine-tuning.
- **Computational Cost:** Full BERT fine-tuning requires GPU access (2-3 hours on CPU). Google Colab or a cloud GPU is recommended for reproducibility.
- **External API Dependency:** The Gemini-based summarization method requires network access, a valid API key, and is subject to third-party rate limits, latency, and per-call cost, none of which apply to the self-contained TF-IDF method. Gemini outputs are also non-deterministic by default (controlled via temperature), so repeated calls on the same review can yield slightly different summaries, which complicates exact reproducibility compared to the deterministic TF-IDF baseline.

8.3 Lessons Learned

The project reinforced that the choice of architecture matters less than the quality of the data pipeline and training regime. Significant time was invested in ensuring the preprocessing pipeline was consistent across training and inference — a common source of performance gaps in deployed NLP systems. Additionally, prompt engineering strategies benefit greatly from domain knowledge: the emotion word lexicon and judgment vocabulary lists used in the audience and critic strategies were curated through careful analysis of the review corpus.

9. Conclusion & Future Work

9.1 Conclusion

This project successfully implemented an end-to-end AI system for movie review sentiment analysis and summarization, applying seven distinct techniques from the Advanced AI course. The fine-tuned DistilBERT model achieves 93%+ accuracy on the IMDb benchmark, and the four-strategy prompt engineering summarizer produces persona-appropriate summaries in real time through a Flask API and HTML interface. All deliverables - notebooks, model weights, API, and frontend - are publicly available on GitHub.

9.2 Future Work

Several directions could extend this project:

- **Formal Comparison of Summarization Methods:** Conduct a systematic ROUGE and human-preference evaluation comparing the TF-IDF extractive method against the Gemini-based generative method across all four prompt strategies, quantifying the fluency/determinism trade-off discussed qualitatively in Section 6.3. A fine-tuned local model (e.g. T5 or BART) could also be benchmarked as a third, offline generative option.
- **Aspect-Based Sentiment Analysis:** Move beyond binary polarity to identify sentiment toward specific aspects (acting, direction, screenplay, soundtrack), providing richer analytics for studio decision-making.
- **Multi-domain Extension:** Adapt the pipeline to other review domains (music, games, restaurants) through zero-shot or few-shot transfer.
- **Long Document Handling:** Implement a sliding-window or hierarchical attention approach to handle reviews exceeding the BERT 128-token limit without truncation loss.
- **User Feedback Loop:** Integrate a rating mechanism in the frontend to collect human preferences across summary styles, enabling reinforcement learning from human feedback (RLHF) to optimize the prompt strategies.

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Required Technique	Implementation	Notebook
NLP & Text Pre-processing	HTML removal, tokenization, stop-word removal, lemmatization	NB 1
Word Embedding Methods	Word2Vec 100-dim trained on IMDB corpus	NB 1
Transformer-Based Models (BERT)	DistilBERT fine-tuned for sentiment classification	NB 2
Transfer Learning / Fine-tuning	DistilBERT pre-trained weights adapted to IMDB	NB 2
Prompt Engineering	4 strategies (concise, audience, critic, CoT) implemented twice: TF-IDF selection rules and Gemini system instructions	NB 3
Chain-of-Thought Prompting	CoT strategy: context $\rightarrow$ evidence $\rightarrow$ conclusion selection	NB 3
In-Context / Few-Shot Learning	Batch verdict generation with example conditioning	NB 3

```
ai-project-movie-review-sentiment-summarize/
├── data/
│   └── IMDB_Dataset.csv (50,000 reviews)
├── models/
│   ├── word2vec_imdb.model
│   └── bert_sentiment/(saved DistilBERT weights)
├── notebooks/
│   ├── 01_data_exploration_preprocessing.ipynb
│   ├── 02_bert_sentiment_classification.ipynb
│   ├── 03_t5_summarization_prompt_engineering.ipynb
│   └── 04_evaluation_baseline_comparison.ipynb
├── app.py (Flask REST API)
├── index.html (Interactive Frontend)
├── .env (GEMINI_API_KEY)
└── README.md
```

Appendix C: API Endpoint Reference

Endpoint	Method	Description	Response Fields
POST /analyze	POST	Analyze a movie review; body accepts method: "tfidf" or "gemini"	sentiment, summaries, word_count, method
GET /health	GET	Check API status	status, device